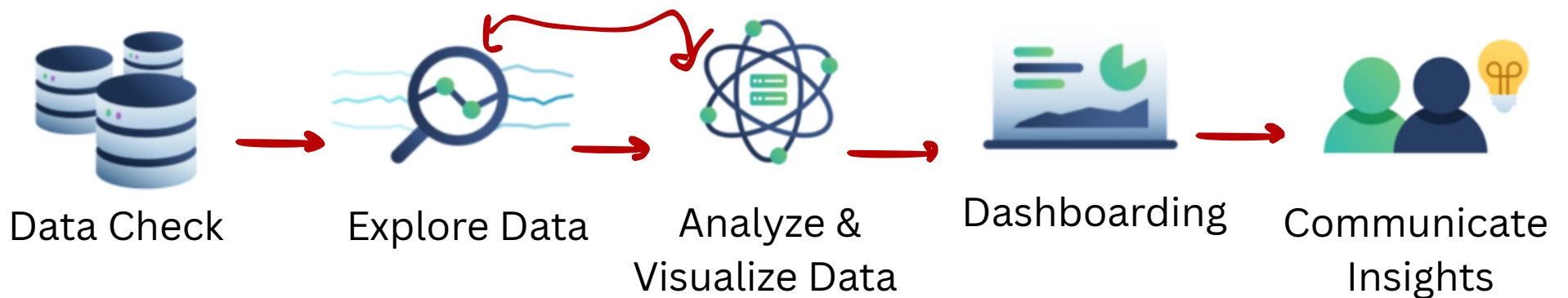


Case Study : Analyzing Customer Churn in Power BI

CHAPTER 1 :

1- Data Analysis Flow in Power BI



Step 1 - Data Check : this step for data cleaning and data validation

- Check for duplicate or missing values
- Do a sense check with other internal data sources

Step 2 - Explore Data : build random charts to understand data

- the better way of explore data is ask yourself the different questions
- Build your first visualizations

Step 3 - Analyze & Visualize data : Actual Visualizations

- Choose the right visualization to convey a message
- Perform more advanced analysis (tracking)

Step 4 - Dashboarding :

- Combine visualization in one or more dashboards

Final Step : Communicate insights :

- Communicate your insights to stakeholders

2- Task or Problem :

Data : A fictitious dataset about churn from (Databel)

Your Task : discover why customers are churning ?



3- What is the churn rate ?

The churn rate , also known as the rate of attrition or customer churn



4- Calculating churn

formula → Churn rate = customers lost / total number of customers

5- The data :

one big table containing 29 columns

One row per customer

PROJECT IN POWER BI

STEP 1 : Data Check

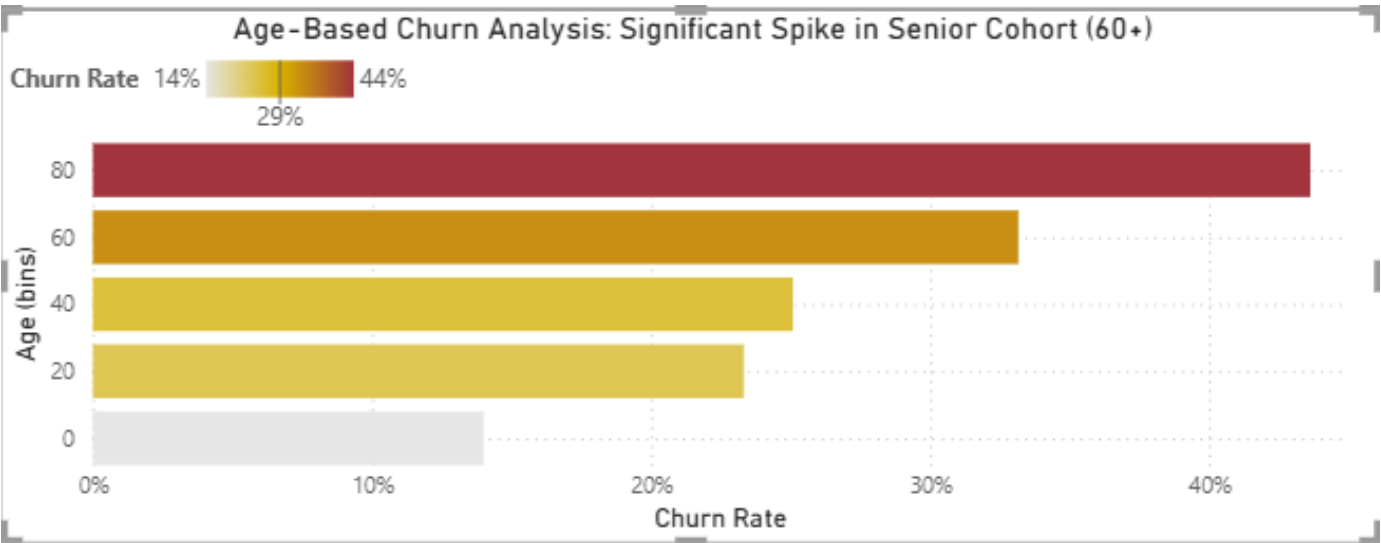
STEP 2 : Explore & Analysis



1- Demographics and Location Analysis:



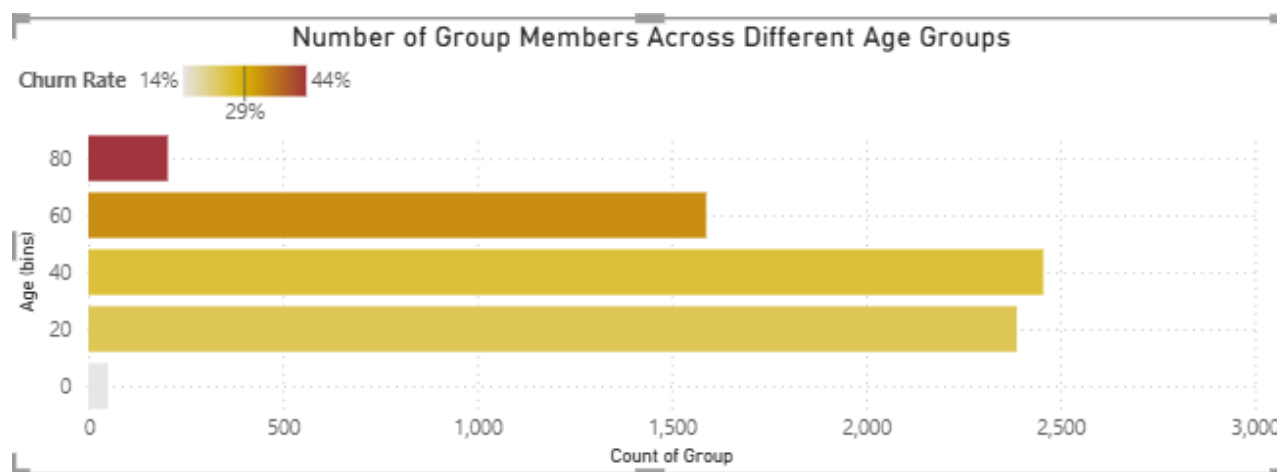
Age-Based Churn Analysis: Significant Spike in Senior Cohort (60+)



Problem : Churn rate increases as customers age, reaching a high of 33% to 44% for those over 60.

Recommendation : Simplify the service to make it easier for older customers, and offer them special discounts or loyalty rewards.

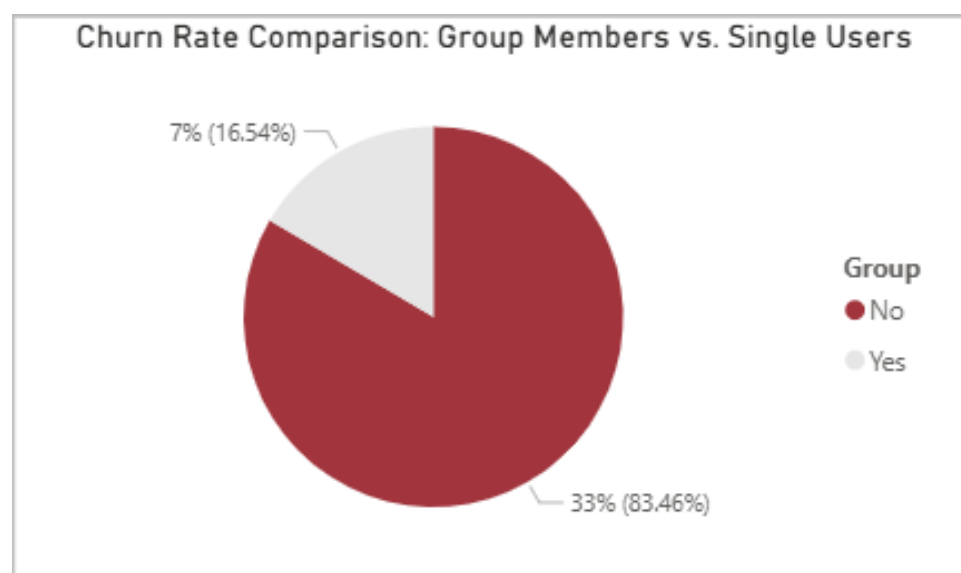
Number of Group Members Across Different Age Groups



Problem : Most customers who participate in "Groups" are between 20 and 60 years old, but as they get older, their participation drops sharply.

Recommendation :Encourage older customers to join or create "Groups" with family members to increase their loyalty and reduce their churn rate.

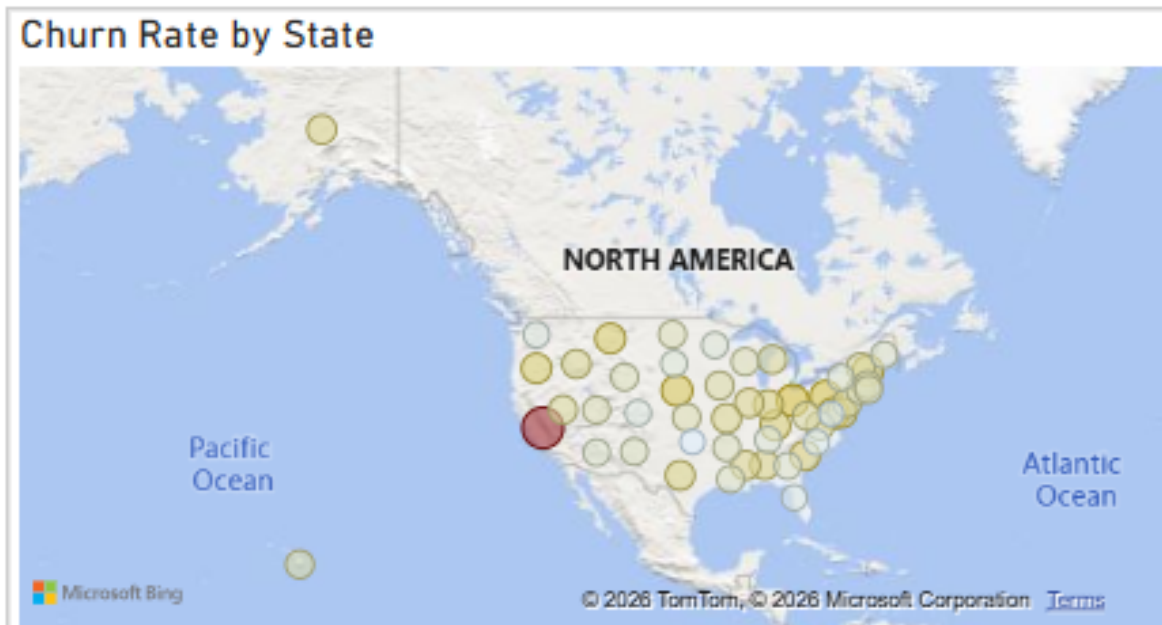
Churn Rate Comparison: Group Members vs. Single Users



Problem :Customers who are not in a "Group" have a much higher churn rate (33%) compared to those who are in a group (7%).

Give single customers special offers or discounts to encourage them to create or join a "Group" with others.

Churn Rate Comparison: Group Members vs. Single Users

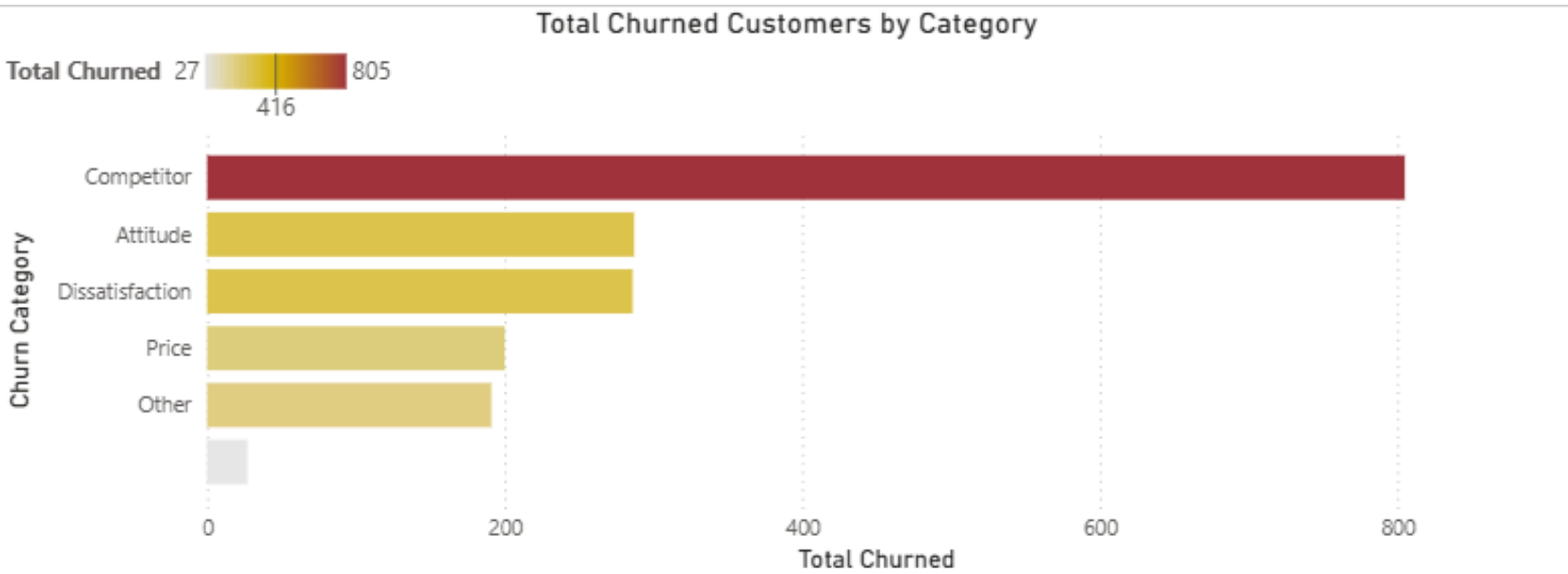


Problem :California (CA) has a critically high churn rate of 63%, making it the biggest problem area compared to all other states.

Recommendation Find out why California has this unique issue, like competitor activities or poor local service, and fix it with a targeted local plan.

2-Root Causes and Warning Signs of Customer Churn

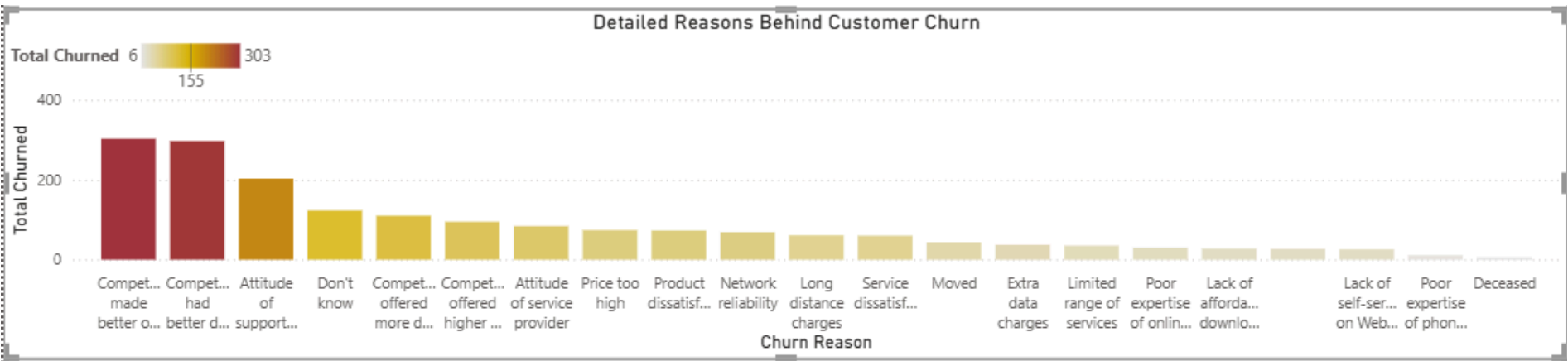
Churn Rate Comparison: Group Members vs. Single Users



Problem :Competitors are the biggest threat, causing over 800 customers to leave

Run a competitive analysis to match or beat rival offers, and launch a loyalty program to make leaving less attractive.

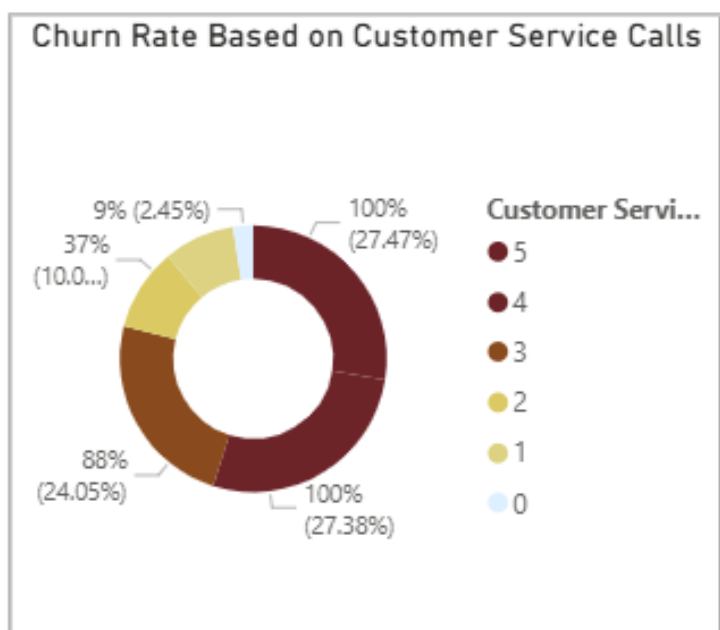
Detailed Reasons Behind Customer Churn



Problem :The two biggest reasons customers leave are competitors offering better options and better devices, both causing around 300 losses each, followed by a negative experience with support staff attitude

Recommendation : Improve our device options and product deals to match what rivals offer, and provide better customer service training to fix staff attitude issues.

Detailed Reasons Behind Customer Churn

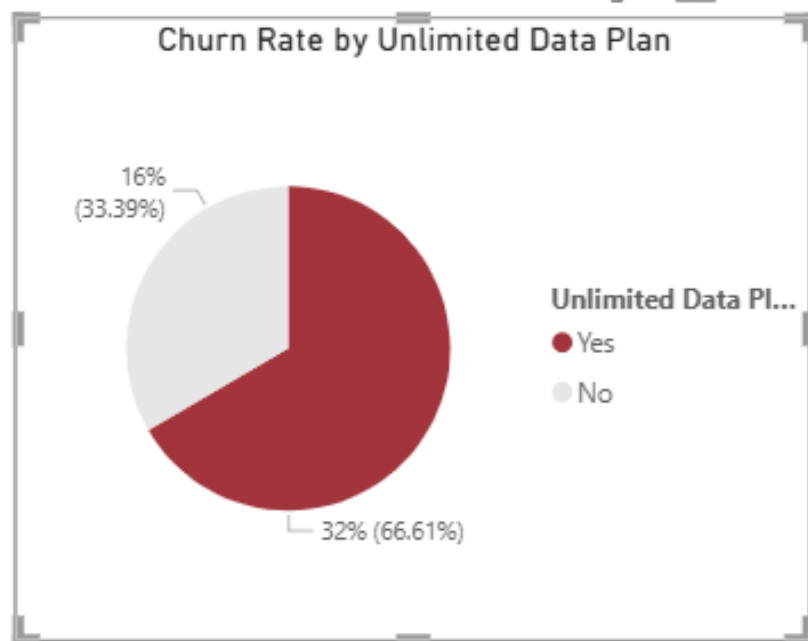


Problem :the churn rate reaches a massive 88% after 3 support calls and hits a guaranteed 100% loss after 4 or 5 calls

Recommendations: > Create a system that flags any customer making a 3rd call, immediately routing them to a senior team authorized to offer instant solutions or refunds

2-Service Plan Analysis

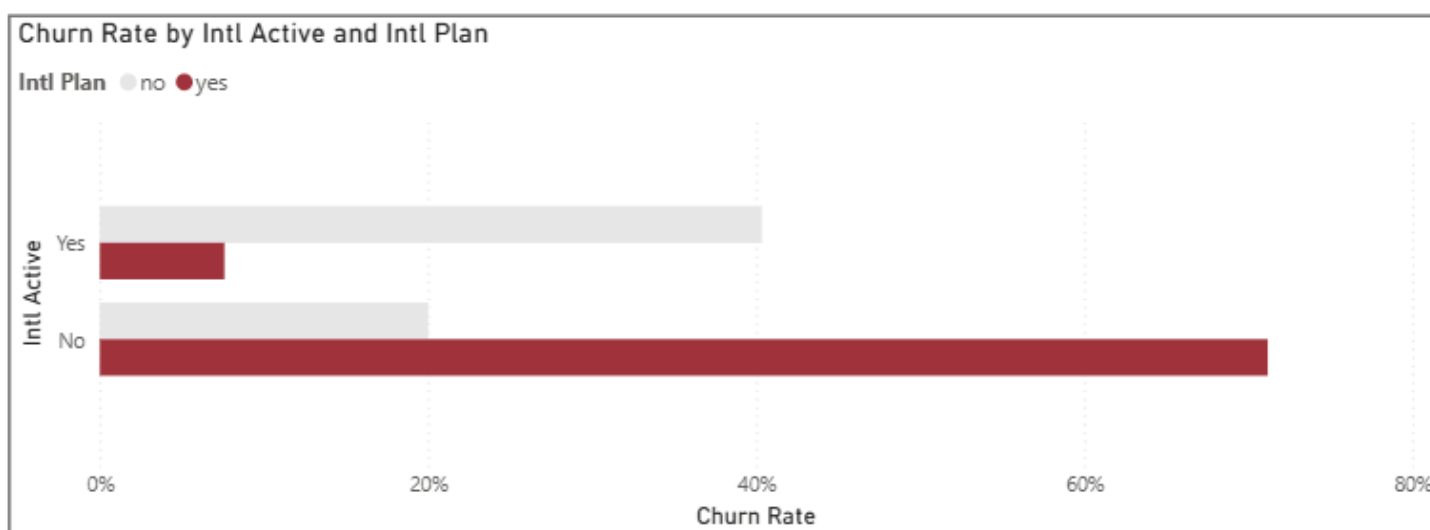
Churn Rate by Unlimited Data Plan



Problem :Customers with an unlimited data plan leave at twice the rate (32%) of those without it (16%), which means the plan is not keeping them loyal

Recommendation :Check if the unlimited data plan is too expensive or has hidden speed limits, and improve its features to make it worth the price

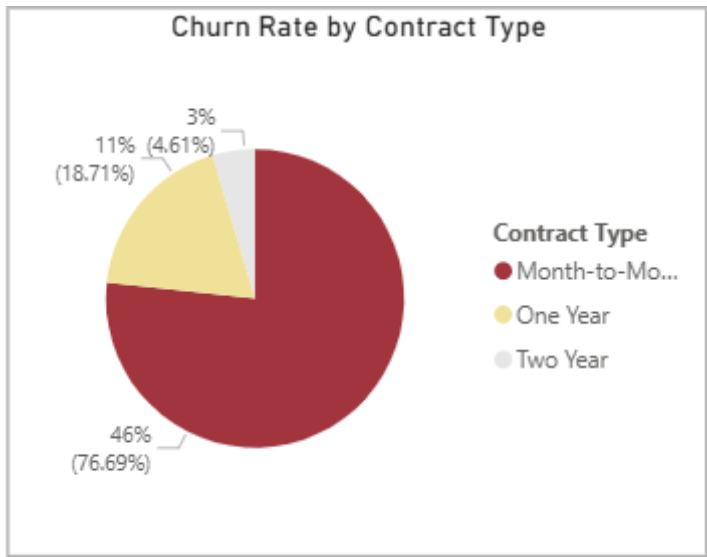
Churn Rate by Unlimited Data Plan



The highest churn rate (over 70%) comes from customers who have an International Plan but are NOT actively using it to make international calls.

Recommendations: > Contact customers who pay for the international plan but don't use it, and offer to switch them to a cheaper or more suitable plan before they change their minds and leave completely.

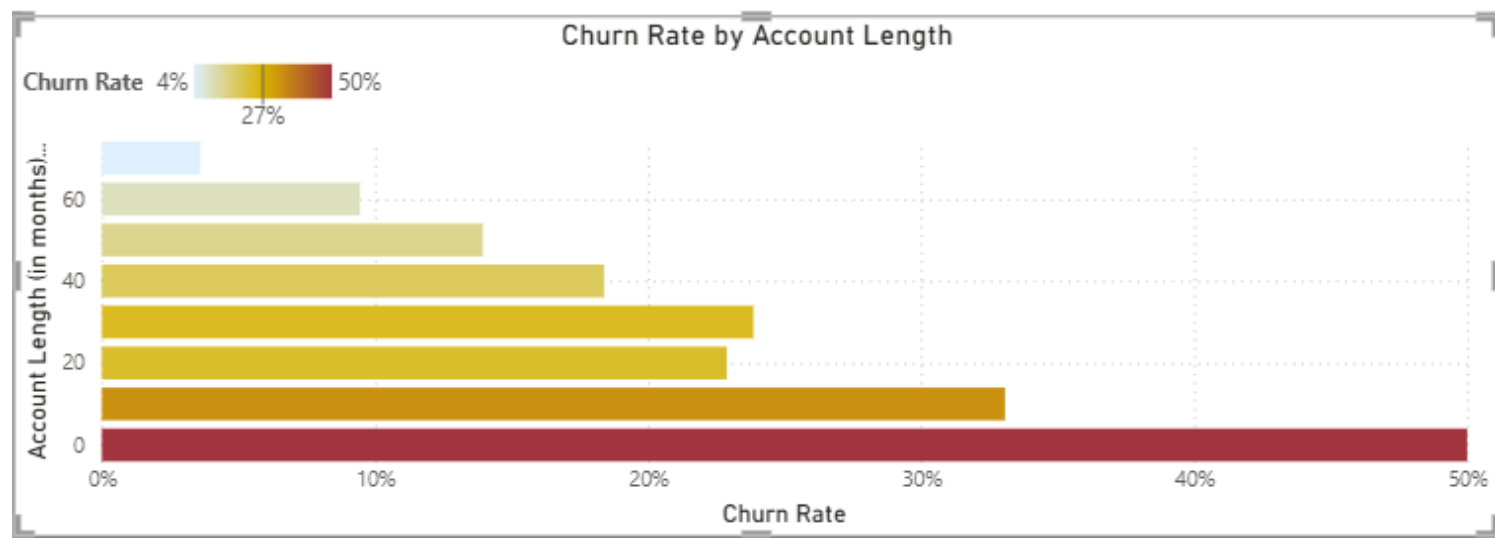
Churn Rate by Contract Type



Problem :Month-to-month contracts are highly unstable, causing a massive 46% churn rate and making up over 76% of all lost customers.

Recommendations: > Give customers discounts or extra benefits if they switch from month-to-month contracts to long-term plans (one or two years).

Churn Rate by Account Length



The Problem: > New customers are the most likely to leave, with the churn rate hitting a dangerous 50% during the very first months (0-10 months) of their subscription.

Recommendations: > Create a welcoming program with special offers and excellent customer care during the first 6 months to help new users get used to the service.